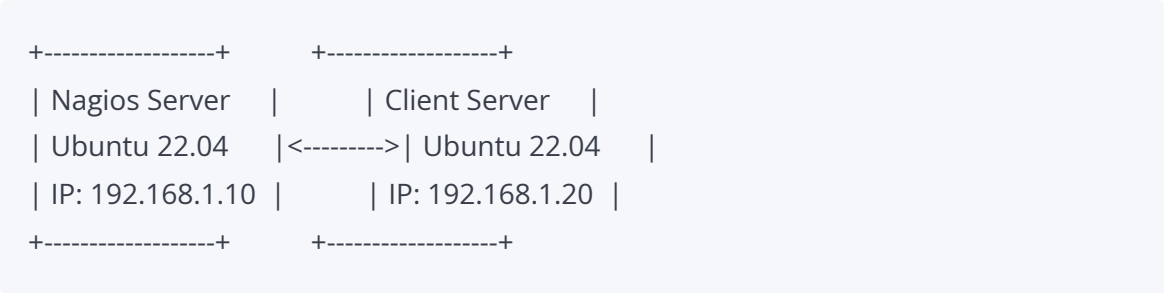


Nagios Monitoring Lab Setup

Lab Topology



Requirements

- 2 Virtual Machines
- Ubuntu Server 22.04 LTS
- 2 GB RAM each
- Network connectivity between VMs

Step 1: Configure Hostnames

Nagios Server

```
sudo hostnamectl set-hostname nagios-server
```

Client Server

```
sudo hostnamectl set-hostname client1
```

Update hosts file on both machines:

```
sudo nano /etc/hosts
```

Add:

```
192.168.1.10 nagios-server  
192.168.1.20 client1
```

Step 2: Install Apache and PHP

On Nagios Server:

```
sudo apt update  
sudo apt install apache2 php libapache2-mod-php wget unzip -y
```

Start services:

```
sudo systemctl enable apache2  
sudo systemctl start apache2
```

Step 3: Install Nagios Core

Download latest Nagios Core from the official website:

```
cd /tmp  
wget https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.9.tar.gz  
tar -xvf nagios-4.5.9.tar.gz  
cd nagios-4.5.9
```

Install dependencies:

```
sudo apt install autoconf gcc make unzip \  
libgd-dev openssl libssl-dev -y
```

Compile:

```
./configure  
make all
```

Create user:

```
sudo make install-groups-users  
sudo usermod -aG nagios www-data
```

Install:

```
sudo make install  
sudo make install-commandmode  
sudo make install-init  
sudo make install-config  
sudo make install-webconf
```

Step 4: Create Nagios Web User

```
sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
```

Set password.

Enable Apache modules:

```
sudo a2enmod rewrite cgi  
sudo systemctl restart apache2
```

Step 5: Install Nagios Plugins

```
cd /tmp  
wget https://nagios-plugins.org/download/nagios-plugins-2.4.12.tar.gz
```

```
tar -xvf nagios-plugins-2.4.12.tar.gz
cd nagios-plugins-2.4.12
```

Compile:

```
./configure
make
sudo make install
```

Step 6: Start Nagios

```
sudo systemctl enable nagios
sudo systemctl start nagios
```

Verify:

```
sudo systemctl status nagios
```

Step 7: Access Web Interface

Open:

```
http://192.168.1.10/nagios
```

Login:

```
Username: nagiosadmin
Password: <your password>
```

Step 8: Install NRPE on Client

On Client VM:

```
sudo apt update  
sudo apt install nagios-nrpe-server nagios-plugins -y
```

Edit:

```
sudo nano /etc/nagios/nrpe.cfg
```

Find:

```
allowed_hosts=127.0.0.1
```

Change:

```
allowed_hosts=127.0.0.1,192.168.1.10
```

Restart:

```
sudo systemctl restart nagios-nrpe-server
```

Enable:

```
sudo systemctl enable nagios-nrpe-server
```

Step 9: Install NRPE Plugin on Nagios Server

```
sudo apt install nagios-nrpe-plugin -y
```

Test:

```
/usr/lib/nagios/plugins/check_nrpe -H 192.168.1.20
```

Expected:

```
NRPE v4.x
```

Step 10: Add Client Host

Create:

```
sudo nano /usr/local/nagios/etc/objects/client1.cfg
```

Add:

```
define host{
use          linux-server
host_name    client1
alias        Ubuntu Client
address      192.168.1.20
}
```

Add service:

```
define service{
use          generic-service
host_name    client1
service_description PING
check_command check_ping!100.0,20%!500.0,60%
}
```

Step 11: Include Configuration

Edit:

```
sudo nano /usr/local/nagios/etc/nagios.cfg
```

Add:

```
cfg_file=/usr/local/nagios/etc/objects/client1.cfg
```

Verify:

```
sudo /usr/local/nagios/bin/nagios \
-v /usr/local/nagios/etc/nagios.cfg
```

Expected:

Things look okay

Restart:

```
sudo systemctl restart nagios
```

Useful Monitoring Services

CPU Usage

```
define service{
  use generic-service
  host_name client1
  service_description CPU Load
  check_command check_nrpe!check_load
}
```

Disk Usage

```
define service{
  use generic-service
  host_name client1
  service_description Disk Usage
```

```
check_command check_nrpe!check_disk
}
```

Memory Usage

```
define service{
use generic-service
host_name client1
service_description Memory
check_command check_nrpe!check_mem
}
```

SSH Service

```
define service{
use generic-service
host_name client1
service_description SSH
check_command check_ssh
}
```

Practicals You Can Perform

1. Host Monitoring
2. CPU Monitoring
3. Memory Monitoring
4. Disk Monitoring
5. SSH Service Monitoring
6. HTTP Service Monitoring
7. Email Alerts
8. Host Down Detection
9. Service Failure Detection
10. Network Monitoring